MILITARY SPECIFICATION

BURSTER, PROJECTILE, T18, LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing of one burster designated as Burster, Projectile, T18.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL–A–2550 —Ammunition and Special Weapons, General Specification for

MIL–R–11470 —Radiographic Inspection: Qualification of Equipment, Operators and Procedures

MIL–R–11471 —Radiographic Inspection of Metals

MIL–I–45208 —Inspection Requirements, General Specification for

MIL–I–45607 —Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL–STD–105D—Sampling Procedures and Tables for Inspection by Attributes

MIL–STD–109 —Quality Assurance Terms and Definitions

MIL–STD–1235 —Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ORDNANCE CORPS

8799216—Carton, Packing, Ammunition, for Charge, Burster, T18E1

8799217—Box, Packing, Ammunition for Charge, Burster, T18E1

8830868—Projectile, Burster, T18 Assembly

PUBLICATIONS

ORDNANCE CORPS

IEL–8830868—Index of Inspection Equipment Lists

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Material.** Materials shall be in accordance with applicable drawings and specifications.

3.2 **Burster.** The burster shall comply with all requirements specified on Drawing (dwg.) 8830868 and with all requirements specified in applicable specifications.

3.3 **Cavitation.**

3.3.1 *Area from the bottom to one inch above bottom.* There shall be no single cavity having a projected length of more than 1/8

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inch or a sum total of cavities of more than 1/64 square inch. There shall be no cracks or annular rings in the charge in this area.

3.3.2 *Area from top to one inch from bottom of container.* There shall be no single cavity having a projected length of more than one inch or a sum total of cavities of more than 3/8 square inch and there shall be no more than three transverse cracks 1/32 inch wide, maximum, in this area.

3.3.3 *Porosity.* Porous areas shall be treated as cavities, and shall be subject to restrictions placed upon cavities, except that 80 percent of the projected length and 80 percent of the projected area shall be considered for acceptance purposes.

3.3.4 *Void between charge and base plug.* There shall be no void greater than 1/32 of an inch between the charge face and the face of the base plug.

3.4 **Workmanship.** All parts shall be free from burs, chips, cracks, dirt, grease, rust and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 **General quality assurance provisions.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Contractor quality assurance system.* The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL–I–45208.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
(b) Number of units of product inspected.
(c) Results obtained for all inspection performed.
(d) Drawing, specification number and date, together with an identification and date of changes.
(e) Certificates of compliance on all material purchased directly by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.
(f) Number of items in the lot.
(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL–I–45208 and will be

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performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of charge burster assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit as determined by the Government shall not necessitate changing the lot number. Product shall be submitted for inspection in accordance with Standard MIL-STD-105D (or Standard MIL-STD-1235, when applicable). Each lot shall contain:

(a) Composition B4 from not more than one lot.

(b) Parts produced by one manufacturer in accordance with the same drawing and same drawing revision.

4.2.2 *Examination.* Sampling plans and procedures for the following classification of defects shall be in accordance with Standard MIL-STD-105 except that inspection for Critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Plug, base (see dwg. 8883596 or 8883597, as applicable, covering a detail of 8830868).

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.2).*
Critical:	None defined		
Major:	None defined		
Minor:	AQL 2.50 percent		
201.	Large diameter, minimum, min.	Gage	01001
202.	Small diameter, min.	Gage	01002
203.	Total length	Gage	01003
204.	Distance to shoulder of large diameter	Gage	01004
205.	Evidence of poor workmanship (see 3.4)	Visual	01005

4.2.2.2 Tube, burster, or tube, burster, alternative prior to painting (see dwg. 8860531 or 8860532, as applicable, covering a detail of 8830868).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined		
Major:	None defined		
Minor:	AQL 1.50 percent		
201.	Total length min.	Gage	02001
202.	Diameter, min.	Gage	02002
203.	Wall thickness	Gage	02003
204.	Evidence of poor workmanship (see 3.4)	Visual	02004

4.2.2.3 Tube, burster, or tube, burster, alternative (see dwg. 8860531 or 8860532, as applicable, covering a detail of 8830868).

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Categories	Defects	Method of inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Interior or exterior surface not completely painted	Visual	03001
102.	Foreign matter in interior	Visual	03002
103.	Pool of paint	Visual	03003
Minor:	None defined		

4.2.2.4 Assembly (see dwg. 8830868).

Categories	Defects	Method of inspection	Code No.
Critical:			
1.	Base plug missing	Visual	04001
2.	Burster tube cracked	Visual	04002
3.	Explosive charge loose in container	Manual	04003
4.	"Up" arrow points in wrong direction	Visual	04004
5.	Closure missing	Visual	04005
Major:	AQL 0.25 percent		
101.	Total length	Gage	04006
Minor:	AQL 1.50 percent		
201.	Closing disc ruptured	Visual	04007
202.	Closing disc or base plug incompletely sealed	Visual	04008
203.	Evidence of poor workmanship (see 3.4)	Visual	04009
204.	Marking misleading or unidentifiable, including incorrect color code	Visual	04010

4.2.2.5 Carton prior to sealing (see dwg. 8799217).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined		
Major:	AQL 0.25 percent		
101.	Burster projectile missing	Visual	05001
Minor:	AQL 0.40 percent		
201.	Any packing component missing	Visual	05002

4.2.2.6 Carton sealed (see dwg. 8799216).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined		
Major:	AQL 0.25 percent		
101.	Tape improperly applied	Visual	06001
Minor:	AQL 1.00 percent		
201.	Marking misleading or unidentifiable	Visual	06002
202.	Contents loose upon shaking	Manual	06003

4.2.2.7 Carton in bag, sealed (see dwg. 8799216).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined		
Major:	AQL 0.25 percent		
101.	Bag improperly sealed, torn or cut	Visual	07001
Minor:	None defined		

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4.2.2.8 Sealed wood packing box (see dwg. 8799217).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined		
Major:	AQL 0.25 percent		
101.	Box damaged	Visual	08001
102.	Strapping missing, broken or loose	Visual-Manual	08002
Minor:	AQL 1.50 percent		
201.	Strapping improperly assembled	Visual	08003
202.	Lid improperly assembled	Visual	08004
203.	Markings misleading or unidentifiable	Visual	08005
204.	Contents loose in box	Manual	08006

4.2.3 *Testing.*

4.2.3.1 Cavitation (see 3.3). Cavitation defects are classified as follows:

Critical code no. 040011
Any cavity, crack or annular ring in the area from the bottom to one inch above the bottom (see 3.3.1). Any void between charge and base plug in excess of $\frac{1}{32}$ inch (see 3.3.4).
Major code no. 04012
Cavitation or cracks in excess of that permitted by 3.3.2 or 3.3.3.

The sampling plan and procedure for this test for both critical and major defects shall be in accordance with Standard MIL–STD–1235 using a CSP–2 plan having an AQL of 0.25 percent except that when a critical defect is found, 100 percent inspection for i successive defect free units (for both criticals and majors) shall be required immediately instead of continuing on a sample basis. When a major defect is found during the f phase of the plan, inspection shall continue at the same frequency until the homogeneity requirement is fulfilled or 100 percent inspection is required. Appropriate i and f values shall be as specified by the procuring activity. The test shall be performed as specified in 4.3.1.

4.2.4 *Inspection equipment.* Index of Inspection Equipment Lists Number IEL 8830868 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL–I–45607.

4.2.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures.

4.3.1 *Determination of cavities.* The sample bursters shall be subjected to x-ray examination. In all exposures the x-ray beam shall

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form a 90 degree angle with the longitudinal axis of the projectile. Examination of the x-ray negative shall be made to determine compliance with applicable requirements.

Note. Radiographic equipment, operations and procedures shall be qualified in accordance with Specification MIL-R-11470. In addition, the sensitivity of x-ray technique shall be 2.0 percent, and the photographic density of the film used shall be no less than 1.25 as determined in accordance with 4.3.1.1 and 4.3.1.2. In the event an original x-ray picture leaves doubt as to acceptability, the burster shall be re-x-rayed at an angle 90 degrees from that of the original exposure and in the same plane.

4.3.1.1 *Check of sensitivity of x-ray technique.* The sensitivity of x-ray technique shall be determined by means of a penetrameter fabricated as specified herein. The penetrameter shall be placed on the side of the burster away from the film (source side). Observation shall be made of the image of the penetrameter on the x-ray film. The sensitivity of x-ray technique shall be considered satisfactory when the image of the penetrameter, including the hole in the center, is sharply defined on the x-ray film. Unless otherwise specified, the penetrameter shall be made of steel having an approximate specific gravity of 7.8. The thickness of the penetrameter shall be based on a combination of both charge and steel thickness and shall be equal to 2.0 percent of the maximum equivalent thickness. The penetrameter shall have a hole, the diameter of which is twice the thickness of the penetrameter. For further details on construction of the penetrameter see Specification MIL-R-11471. The actual thicknesses may be determined from a radiograph of the metal parts. The penetrameter thickness shall be determined as illustrated in figure 1 of this specification.

4.3.1.2 *Check of photographic density of film.* The photographic density of the film shall be determined on the film in a region of sound charge plus metal corresponding to that portion of the material which is of the greatest combined thickness to be examined on the negative. To judge densities, it is recommended that standards of radiographic densities be used (see 6.4). Comparison of film with standards may be made visually; however the use of a densitometer is recommended.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level C.* The T18 projectile burster shall be preserved and packaged in accordance with dwg. 8799216.

5.2 Packing.

5.2.1 *Level C.* The cartons shall be packed in accordance with dwg. 8799217.

5.3 Marking. The cartons and boxes shall be marked in accordance with dwg. 8799216 or 8799217 as applicable.

5.4 Data cards. Data cards shall be prepared for each lot in accordance with the information specified in Specification MIL-A-2550.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (00 Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to

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prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "contractor design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5, and 6.3.6 will be designated in the inspection equipment list by the following codes:

CDOF —Contractor's design responsibility on Ordnance format; ORDM–608–12.
CDCF —Contractor's design responsibility on contractor format.
OD —Ordnance design.
ODMU —Ordnance design, mandatory for use.
CE —Commercial equipment.
SCD —Specification Control Drawing.

6.3.3 *Ordnance designs.* Ordnance designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and nonmandatory. When an inspection equipment list references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with nonmandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawings.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDOF shall comply with the format and requirements of ORDM 608–12, and, in addition, contain the detail and information specified above.

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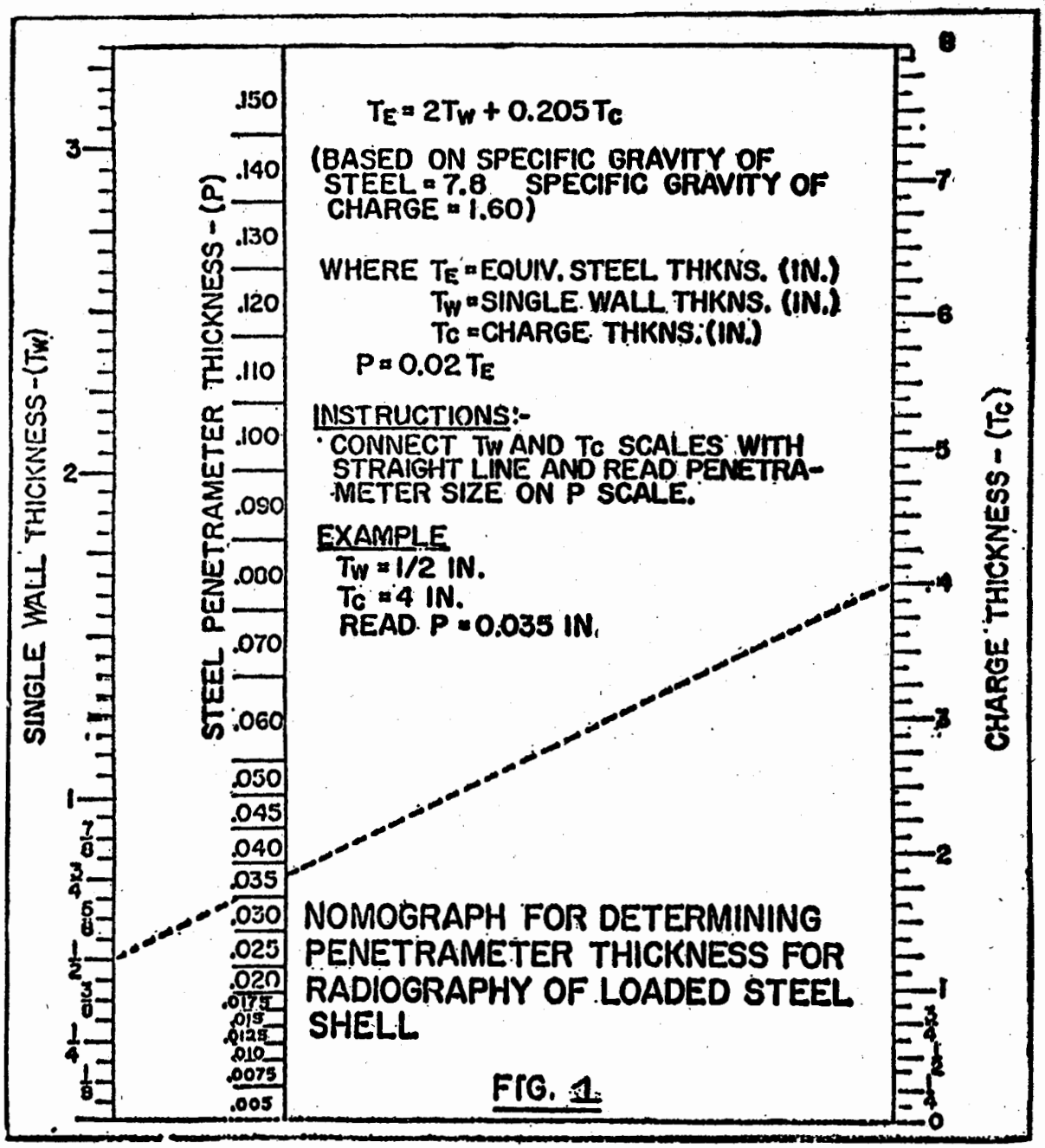
6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 *Film density.* "American Standard for Diffuse Transmission Density" published by American Standards Association, 70 East 40th Street, New York, N. Y., has been found satisfactory as a source of information on film densities.

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NOMOGRAPH FOR DETERMINING PENETRAMETER THICKNESS FOR RADIOGRAPHY OF LOADED STEEL SHELL

FIG. 1.

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